

Lick Kast Polarimetry — Exposure Time & S/N Plan

UV-excess QSO candidates (34 targets with g photometry)

Methodology (K. Leighly notes 2026-07; Chromey, "To Measure the Sky", sec. 9.5.4)

1. Each target is run through the Kast ETC (https://etc.ucolick.org/web_s2n/kast) with a flat template and its PS1 g-band AB magnitude + 0.752 mag: the polarization optics pass half the flux ($m_{\text{pol}} = m + 2.5 \log 2$).
2. ETC settings, identical for all targets: dichroic d55, slit 2.0 arcsec, binning 1x1 pixels, grism 600/4310, grating 600/7500, seeing 1.5", airmass 1.1, reference exposure 900 s.
3. The ETC returns per-pixel object counts, sky counts, and read noise vs. wavelength. Spectropolarimetry uses the per-pixel S/N of the median pixel in 4000-5500 Å; imaging polarimetry integrates the counts over the PS1 g profile (synphot) and applies the CCD equation $S/N = N_{\text{qso}} / \sqrt{N_{\text{qso}} + N_{\text{sky}} + \text{Noise}}$.
4. The exposure time reaching the required S/N is the exact solution of the CCD equation (Chromey eq. 9.77). Required S/N = 10 (advisor: "ideally >10"); the 5 floor ("definitely >5") is tabulated in `kast_obs_plan.csv`.
5. Polarimetry penalty: to be sensitive to polarization fraction P, the normal exposure time is multiplied by 1/P (rough rule per notes) – x25 for P = 4%, x100 for P = 1%. Note P is NOT the scattered fraction: scattering with unfavorable geometry can cancel to zero net polarization.

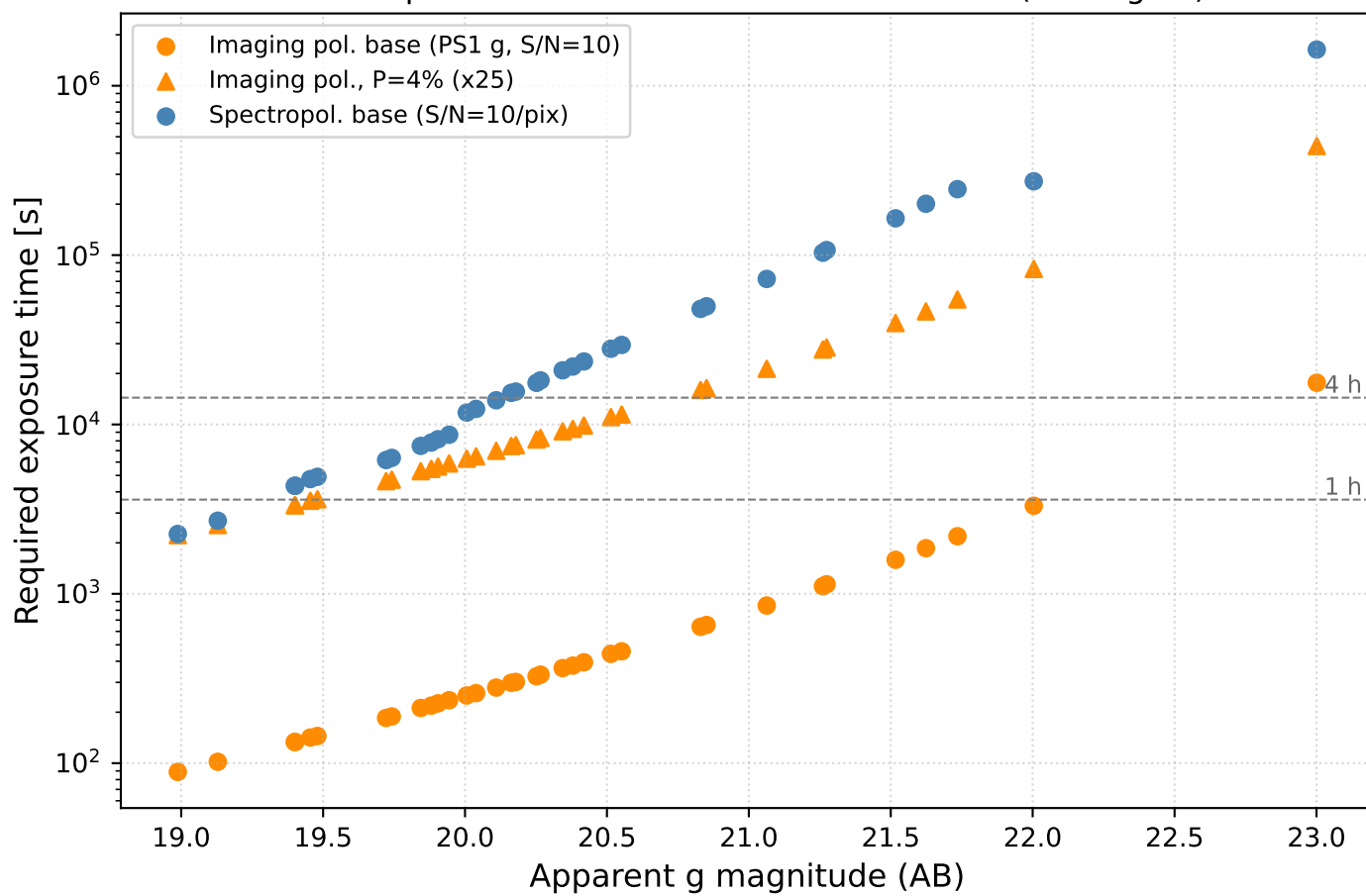
Sample summary at S/N = 10, imaging polarimetry (PS1 g)

P = 4%: 5 targets within 1 h, 15 within 2 h, 24 within 4 h
P = 1%: 0 targets within 2 h, 5 within 4 h, 15 within 8 h

Spectropolarimetry at S/N = 10/pixel is feasible unscaled for the brightest targets (2 within 1 h) but no target fits 4 h once the 1/P factor is applied – imaging polarimetry first, spectropolarimetry only for bright detections, is the natural protocol.

Caveats: g magnitudes are observed-frame, not corrected for Galactic extinction; the 1/P rule is approximate (a 3-sigma detection criterion via $\sigma_P = \sqrt{2}/(S/N)$ scales as $1/P^2$) – confirm convention before the proposal; several candidates lie at $z < 0.5$ (W2M rows) – vet before targeting.

Kast exposure times — UV-excess candidates (34 targets)



Per-target exposure times at S/N = 10 (brightest first; imaging = PS1 g band)

Target	g (AB)	ETC mag	Img base	Img P=4%	Img P=1%	N exp (4%)	Spec base	Spec P=4%
J204626.10+002337.6	18.99	19.74	89 s	37.0 m	2.5 h	2	37.6 m	15.7 h
J150746.11+093731.2	19.13	19.88	102 s	42.5 m	2.8 h	2	45.0 m	18.8 h
TARGETID_39628512285426154	19.40	20.15	133 s	55.6 m	3.7 h	2	72.4 m	30.2 h
TARGETID_39633384095352378	19.40	20.15	133 s	55.6 m	3.7 h	2	72.4 m	30.2 h
J150356.69+125914.3	19.45	20.21	142 s	59.1 m	3.9 h	2	79.3 m	33.1 h
J144809.53+054019.2	19.48	20.23	145 s	60.3 m	4.0 h	3	81.8 m	34.1 h
TARGETID_39628516710418961	19.72	20.47	3.1 m	77.1 m	5.1 h	3	1.7 h	42.8 h
J150357.14+041856.3	19.74	20.49	3.1 m	78.7 m	5.2 h	3	1.8 h	44.1 h
J141354.73+110047.5r	19.84	20.60	3.5 m	88.3 m	5.9 h	3	2.1 h	51.9 h
J152259.51+032347.6r	19.88	20.63	3.6 m	1.5 h	6.1 h	4	2.2 h	54.3 h
J143027.14+152514.8	19.91	20.66	3.8 m	1.6 h	6.3 h	4	2.3 h	56.8 h
J134039.67+051419.7	19.94	20.70	3.9 m	1.6 h	6.5 h	4	2.4 h	60.4 h
J150152.94+020604.9	20.01	20.76	4.2 m	1.7 h	7.0 h	4	3.3 h	81.6 h
J115449.31+112705.8	20.04	20.79	4.3 m	1.8 h	7.2 h	4	3.4 h	85.8 h
TARGETID_39633426973720698	20.11	20.86	4.7 m	1.9 h	7.8 h	4	3.9 h	96.5 h
J110018.80+080455.3	20.16	20.92	5.0 m	2.1 h	8.3 h	5	4.3 h	106.8 h
J132210.17+084232.9	20.18	20.93	5.0 m	2.1 h	8.4 h	5	4.3 h	108.6 h
TARGETID_39633267170738647	20.25	21.00	5.4 m	2.3 h	9.1 h	5	4.9 h	122.3 h
J150013.42+123645.2	20.27	21.02	5.6 m	2.3 h	9.3 h	5	5.1 h	126.5 h
TARGETID_39633136459448983	20.34	21.10	6.1 m	2.5 h	10.1 h	6	5.8 h	145.0 h
TARGETID_39633149587623462	20.38	21.13	6.3 m	2.6 h	10.5 h	6	6.1 h	152.7 h
TARGETID_39633325387681000	20.42	21.17	6.6 m	2.7 h	11.0 h	6	6.5 h	163.6 h
TARGETID_39633396137198490	20.51	21.27	7.4 m	3.1 h	12.3 h	7	7.8 h	194.4 h
TARGETID_39633432401152732	20.55	21.30	7.6 m	3.2 h	12.7 h	7	8.2 h	204.8 h
TARGETID_39633325958105931	20.83	21.58	10.6 m	4.4 h	17.7 h	9	13.4 h	334.3 h
TARGETID_39633162619324602	20.85	21.60	10.9 m	4.5 h	18.2 h	10	13.9 h	346.3 h
TARGETID_39633387157195822	21.06	21.81	14.2 m	5.9 h	23.7 h	12	20.1 h	502.3 h
TARGETID_39632946793025886	21.26	22.01	18.5 m	7.7 h	30.8 h	16	28.7 h	717.7 h
TARGETID_39633311366122701	21.27	22.03	19.0 m	7.9 h	31.6 h	16	29.8 h	743.9 h
J140658.28+161142.6	21.52	22.27	26.5 m	11.0 h	44.1 h	23	45.8 h	1144.7 h
TARGETID_39628379145637914	21.62	22.38	31.0 m	12.9 h	51.7 h	26	55.8 h	1395.9 h
TARGETID_39628346081936841	21.73	22.49	36.5 m	15.2 h	60.8 h	31	68.1 h	1703.0 h
TARGETID_39633085267968176	22.00	22.76	55.2 m	23.0 h	92.0 h	47	75.8 h	1894.7 h
J143404.58+093528.5	23.00	23.75	4.9 h	122.5 h	489.9 h	245	454.3 h	11358.0 h